

# Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and Deepfake Image Detection

R26-IT-011

# Project Supervision

**Supervisor**



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# Team Members



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# Impact of the problem

- Privacy Risks
- Trust and Ethical Concerns
- Limited Accessibility to facial analysis technologies
- Reduced Efficiency in systems

# Existing Solutions & GAP

- Most solutions require cloud-based processing or expensive SDK licenses.
- Many systems depend on high-performance hardware or server infrastructure.(Visage SDK)
- Limited availability of lightweight browser-based solutions accessible to general users.

# The Organizations and the Institutions which faced the problems in absence of this platform

- ***Researchers and Academic institutions***
  - *Many advanced systems require expensive infrastructure or are difficult to integrate into research environments.*

- ***AI and Technology organizations***
  - *Organizations deploying facial AI systems face challenges related to bias, transparency, and deepfake threats.*

- ***Media and Security platforms***
  - *Digital platforms face increasing risks from manipulated facial media and identity misuse.*

# How can this system be beneficial for the organizations and institutions

- ***Researchers and Academic institutions***
  - *Academic institutions can use the system for research, experimentation, and evaluation of demographic estimation models and fairness analysis techniques.*

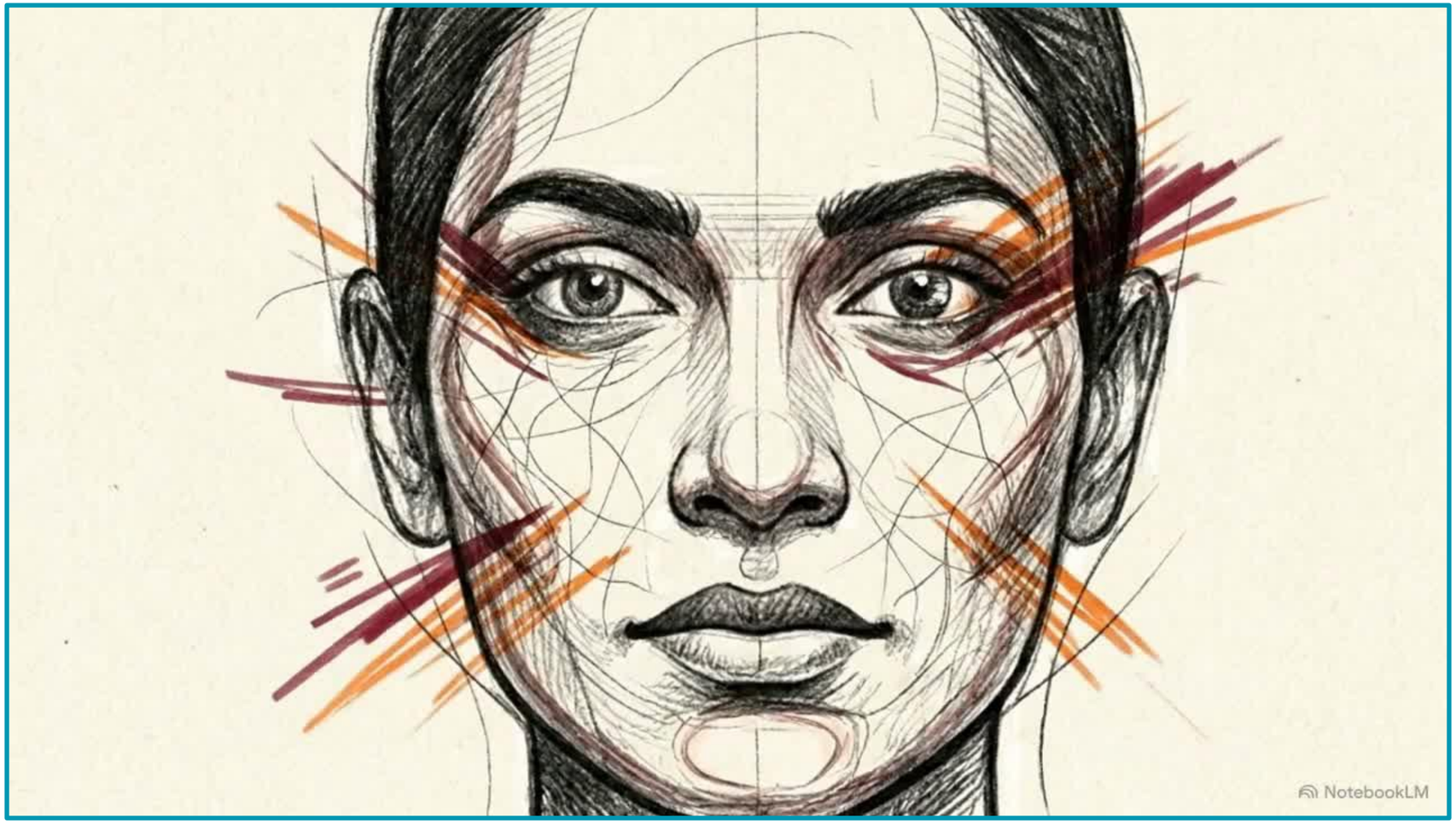
- ***AI and Technology organizations***
  - *This solution supports responsible AI evaluation and facial authenticity verification. helps evaluate prediction reliability and supports more responsible and ethical AI deployment.*

- ***Media and Security platforms***
  - *The system helps detect deepfakes and improve trust, verification, and digital security.*

# PROPOSAL FEEDBACKS & IMPROVEMENTS

| Student ID | Comment | Component | Solution |
|---|---|---|---|
| **IT22541970** | *research value present in the given component. howevert, presentation slides contain certain points which are not able to be explained by the student | **Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition** | Created datasets to compare model performance between Western and South Asian data. |
| **IT22254252** | *inconsistent ideas when speaking about what the scope of the component is. | **Age and Gender Detection using Face-API** | Upgraded the system with a clearly defined scope: real-time browser-based demographic estimation using MobileNetV2 and a fixed dataset pipeline. |
| **IT22278876** | *commercial value or research value of the proposed solution not clearly communciated; if at all exists | **Facial Marks Detection** | evolved this component into a Criminal Face Mark Identification System |
| **IT22276728** | *need some improvement in the application and finding the use cases of the system | **Deepfake Image Detection** | Improved the system and a manually cerated South sri lanka faces dataset, while defining practical use cases such as verify digital identities |

# Video Description




**R26-IT-011**

IT22254252 | PERERA H K G Y

# Age and Gender Detection using Face-API






Specialization:Information Technology

# Age and Gender Detection using Face-API

## Problem Definition

- Most facial analysis systems rely on cloud/server-based processing
- Network latency affects real-time performance
- Need for a lightweight browser-based AI solution

## Research Objective

- Develop a web-based system to detect faces and estimate age and gender in real time

## Research Methodology

- Uses concepts from Artificial Intelligence and Computer Vision
- Facial feature extraction using Face-Api Js
- Real-time image processing from webcam video frames

## Technologies

- Face-api.js for face detection and demographic prediction
- TensorFlow.js for running deep learning models in the browser
- TypeScript for web-based implementation

# Age and Gender Detection using Face-API

## Design Excellence and Contribution

- Successfully structured the Models which are required.
- Successful implementation of the frontend
- Successful implementation of Face-API and Tensorflow.
- Completion of report generation in order to generate a report.

# Age and Gender Detection using Face-API

## Current Situation of the Component




**HOW IT PERFORMS**

**TECHNOLOGIES WHICH WERE USED**

- **NEXT JS**
- **@vladmandic/face-api v1.7.15 -TENSORFLOW JS**
- **Supabase (currently in local storage)**

- **MODELS**
- **TinyFaceDetector**
- **FaceLandmark68TinyNet**
- **AgeGenderNet**
- **Custom MobileNetV2**

- **DATASETS**
- **IMDB-WIKI - AgeGenderNet**
- **ImageNet - Pretrained MobileNetV2**
- **aibloy fairface - Custom MobileNetV2**
- **300-W/HELEN - Landmark68TinyNet**

IT22541970 | Dissanayake DMDN

# Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition






**Specialization:** Information Technology

# Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition

- ***Problem Definition***
  - People from South Asia (like Sri Lanka and India) have different facial features and skin tones compared to people from the West. Because AI didn't trained on enough South Asian faces during its "training," it doesn't understand our expressions as well.

- ***Solution***
  - **Fine-Tuning**: Implementing transfer learning on a custom SSDMoblenet(v3) using localized South Asian data clusters vs. Western benchmarks.
  - **Result**: Quantifying the "Demographic Bias" and providing a localized optimization model that recovers lost accuracy for Sri Lankan/Indian phenotypes.

# Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition

- ***User Requirements addressed by the solution***
  - Ensuring the system recognizes South Asian expressions with the same precision as Western faces.
  - Optimizing the SSDMobilenet backbone to run in browser environments with inference speeds under 50ms.

- ***User Feedback on Prototype***
  - **Supervisor Feedback**: Refined the SSDMobilenet architecture to improve classification accuracy for "Neutral" vs. "Sad" expressions in South Asian clusters.

- ***Design Excellence***
  - **Dual-Stream Benchmarking Architecture:** Designed a system that can compare two different data clusters (Western vs. South Asian) at the same time to identify accuracy gaps.

IT22278876 | Deshan V.W.S

# Facial Marks Detection






Specialization:Information Technology

# Facial Marks Detection

- ***The Critique***
  - commercial value or research value of the proposed solution not clearly communciated; if at all exists
- ***The Strategic Response***
  - Repositioning technology as a Forensic Tool for law enforcement where self diagnosis is irrelevant and automated precision is mandatory.
- **Research Significance**
  - Utilizing permanent facial markers (scars, moles, bruise) as "Soft Biometrics" to identify suspects when primary Identification fails.

# Facial Marks Detection

- **Digital Evidence**

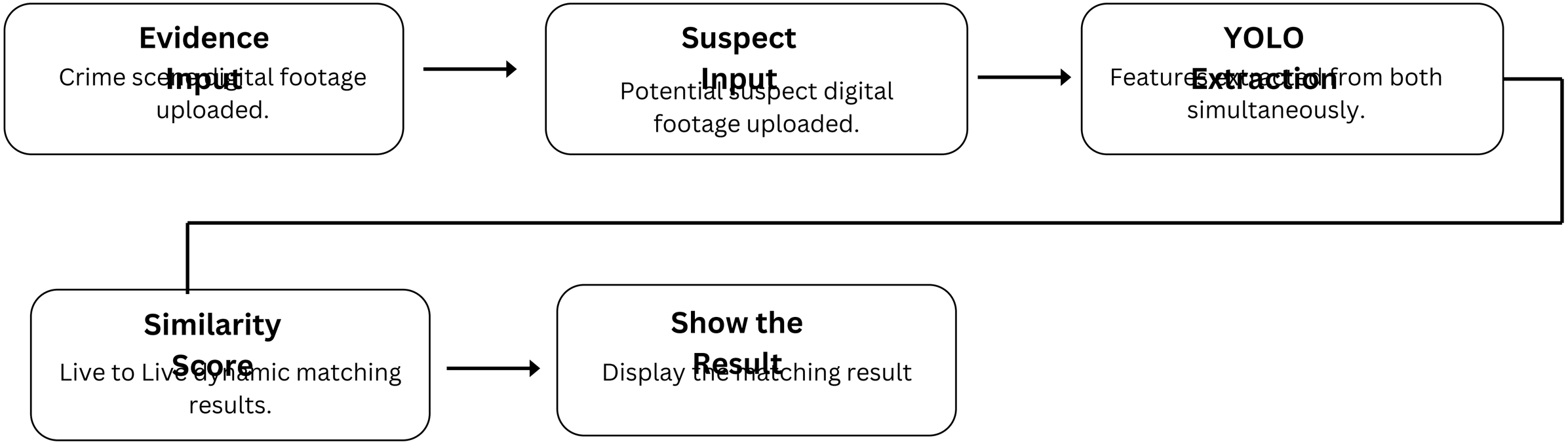
  Instead of low res CCTV, mobile and digital camera footage provide the resolution needed for Micro feature Detection.
  Standard systems fail in low lighting or when faces are partially covered. Permanent markers (scars/moles) are often the only visible evidence.

  - Mobile Footage
  - Digital Still Images
  - Public Sourced Evidence

- **Technology**
  - Higher Mean Average Precision
  - Low Computational Overhead (Colab Efficient)
  - YOLOv11

# Facial Marks Detection

- **Dynamic Workflow Process (**No static database required. Real time ad hoc matching ensures forensic flexibility**.)**



IT22276728 | Gunawardhana W.M.R.T

# Deepfake Image Detection




**Specialization**:Information Technology



- **How the Solution Addresses the Sub-Problem / Prototype**

### 1. Sub-Problem

- The inability of the human eye and traditional image analysis tools to distinguish between authentic photographs and **high-fidelity "Deepfakes"** generated by advanced AI. This leads to the rapid spread of misinformation and the erosion of digital trust.

### 1. Solution

- An advanced neural framework utilizing **Swin-Base** architectures. This solution analyzes both global context and hierarchical features to accurately classify images as real or synthetically generated.

- **User Requirements Addressed**

  - **High Accuracy**
  - **Efficiency**
  - **User-Friendly Interface**
  - **Data Security**

- **Design Excellence / Contribution**
  - **State-of-the-art Technology:** Shifting from traditional CNNs to more robust Transformer-based models for better generalization
  - **Forensic Explainability:** Providing visual cues (such as heatmaps) to justify the detection results, making the AI's decision-making transparent.
  - **Validation Performance:** The model demonstrated exceptional accuracy and stability during testing on a dataset of 2000 diverse images.

- **User Feedback**

  **Supervisor Feedback:** Addressed a significant research gap by developing a localized deepfake detection framework, specifically optimized for Sri Lankan facial demographics where existing global models often underperform

- ***Commercialization***

  - Selling the localized emotion recognition engine as an API for South Asian software developers.
  - Providing continuous model updates and accuracy audits for enterprise clients.

- ***Customer Persona***

  - In retail and banking who need to track local customer satisfaction accurately.
  - Building localized social media or mental health apps for the South Asian market.

- ***How will the cost be recovered***

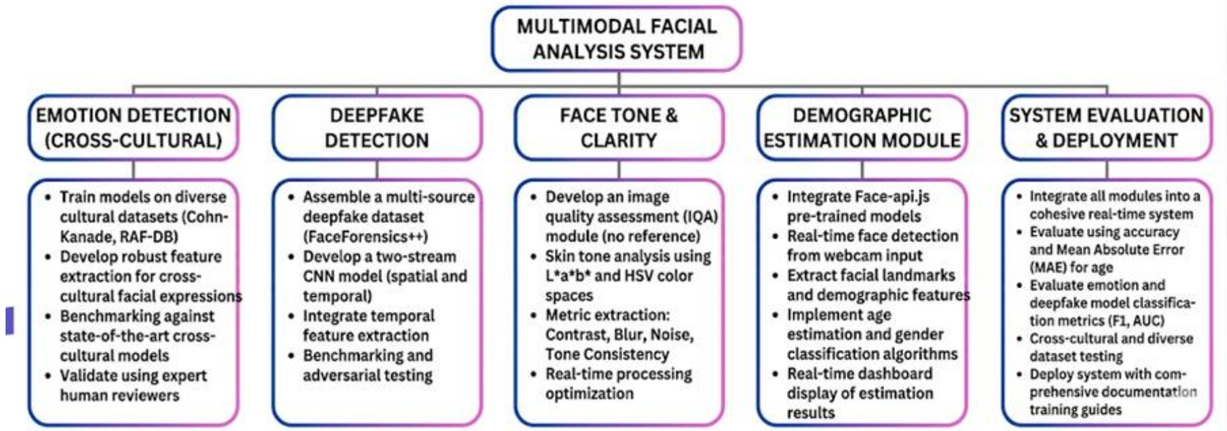
  - Charging for API calls or a monthly fixed fee for business integration.

# Commercialization/Sustainability

## GANT CHART

| Phase | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct |
|---|---|---|---|---|---|---|---|---|
| Literature Review & Benchmarking | | | | | | | | |
| AI Model Development | | | | | | | | |
| Image Preprocessing & Testing | | | | | | | | |
| UI/UX Design | | | | | | | | |
| App Development | | | | | | | | |
| Face-API testing and Lightweight model testing | | | | | | | | |
| Final Validation and PWA activation | | | | | | | | |

## WORK BREAKDOWN STRUCTURE



# Commercialization/Sustainability

## Commercialization / Sustainability

**API-as-a-Service:**
License the localized emotion recognition & deepfake detection engine as a SaaS API for South Asian developers and enterprises.

**Continuous Model Updates:**
Subscription model providing quarterly accuracy audits and retraining on new demographic datasets.

**Academic Licensing:**
Partner with SLIIT and regional universities as a research tool platform for facial AI fairness studies.

## How Costs Will Be Recovered

**Pay-per-API-call:**
Charge per 1,000 API calls for business integration (Emotion/Deepfake endpoints).

**Monthly SaaS Tier:**
Fixed monthly fee covering unlimited in-browser requests for SME clients.

**Enterprise License:**
Annual contract for large-scale deployments in banking, telecom, or government.

**Grant Funding:**
Apply for SLIIT Research Grants and ICTA Sri Lanka Digital Innovation grants.

## Customer Persona

**Retail & Banking Firms**
Need accurate local customer emotion & demographic data for CX analytics.

**Media Platforms**
Detect deepfakes in user-uploaded content and digital identity verification workflows.

**AI Research Institutes**
evaluation of demographic estimation models and fairness analysis techniques.

# Commercialization

| Item | Estimat (LKR) |
| --- | --- |
| Software Tools (Face-api.js, TensorFlow.js, VS Code, Git) | |
| Development Environment (Laptop / Desktop with Webcam) | |
| Internet Access | |
| Testing Environment | |
| Documentation and Report Preparation | |

**ESTIMATED BUDGET**

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GitHub repository: https://github.com/justadudewhohacks/face-api.js
— Provides pre-trained models for age and gender estimation that can run directly in web browsers.

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— Framework for running deep learning models like CNNs in the browser, enabling age and gender detection on client-side.

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# Thank You&Q&As



